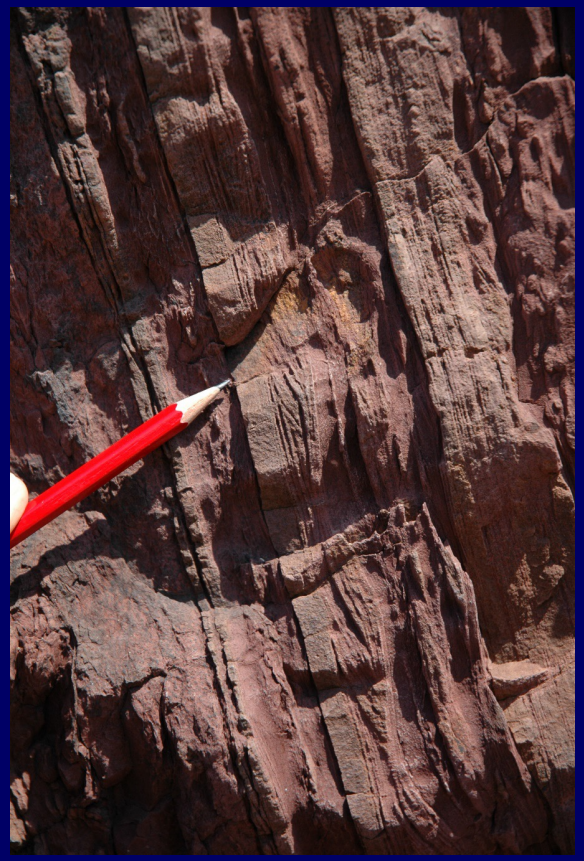
Class 2 Physics of Sedimentation

A fluid dynamics field trip

- The nature of fluids and flows
- Why do fluids move?
- Channel flows
  - Laminar vs turbulent flows
  - Flow Reynold's number
  - Vertical velocity profiles and 'no-slip condition'
  - Basic structure of turbulent flows
- Basics of sediment transport
  - Forces on particles
- Significance of turbulence for erosion and sedimentation



Platte River, Nebraska



Practical

The role of particle settling in sediment transport and deposition

- Forces on particles in static fluids
- Stokes' Law

The physical properties of fluids

Fluids: air and water

Density (ρ)



mass of fluid per unit volume of fluid

Density of water =1 kg/litre or 1000kg/m^3 @10°C
Density of air ~1.3 kg/m^3 (at sea level and room temperature)

Viscosity (μ)

$$\mu = \frac{F}{(A \times (\frac{dU}{dy})}$$

a measure of the resistance of the fluid to deformation

- molecular or dynamic viscosity
- applies not only to fluids but also to plastics, rubber, and other things that can "flow" or deform

Specific weight (γ)

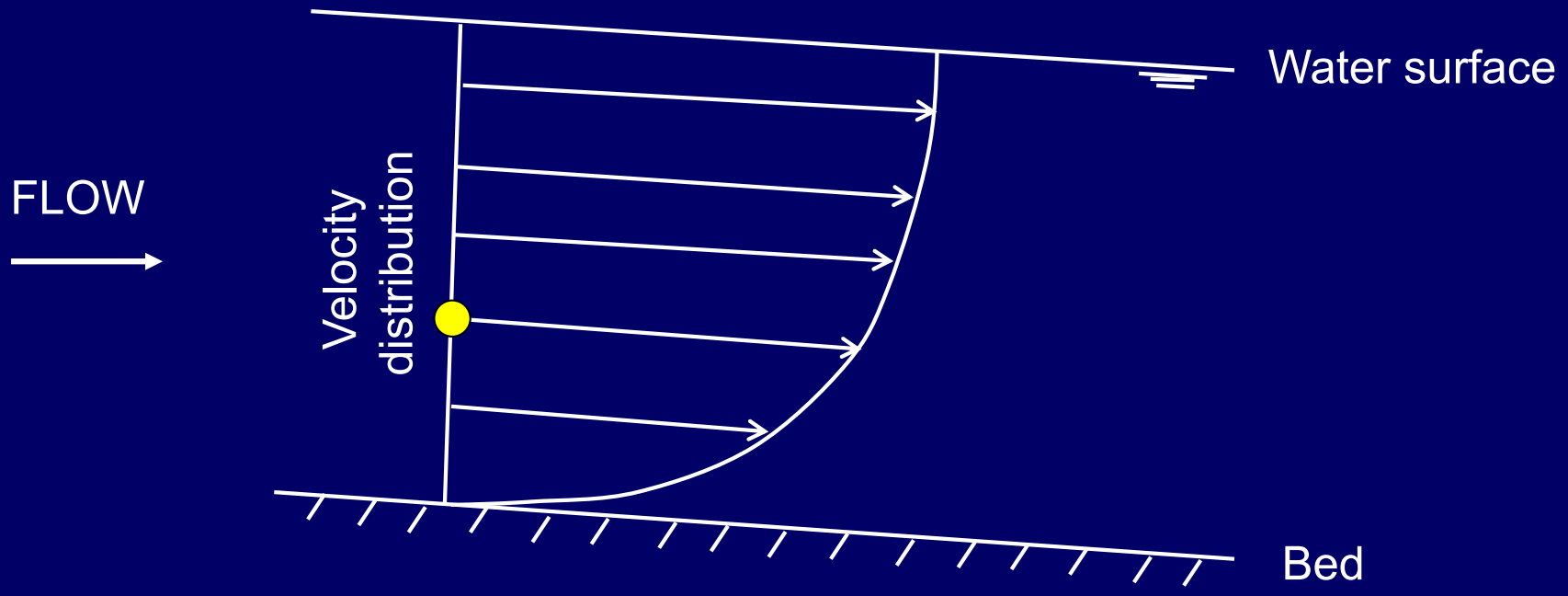
the weight of the fluid per unit volume of fluid

Variables of flows

Ways to describe a flow:

- velocity of the flow
- force per unit area the flow exerts on its boundary (boundary shear stress τ_0)

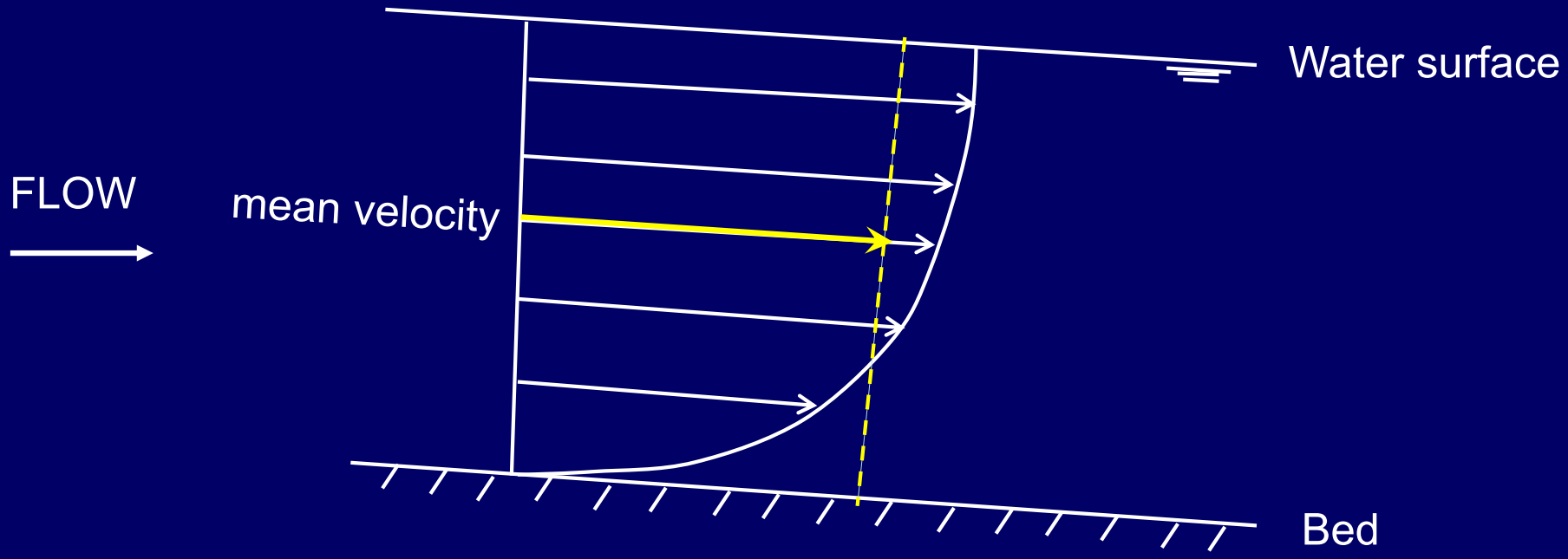
Velocity



Time-averaged local velocity varies continuously from point to point all over the cross section of the flow

Variables of flows: Velocity

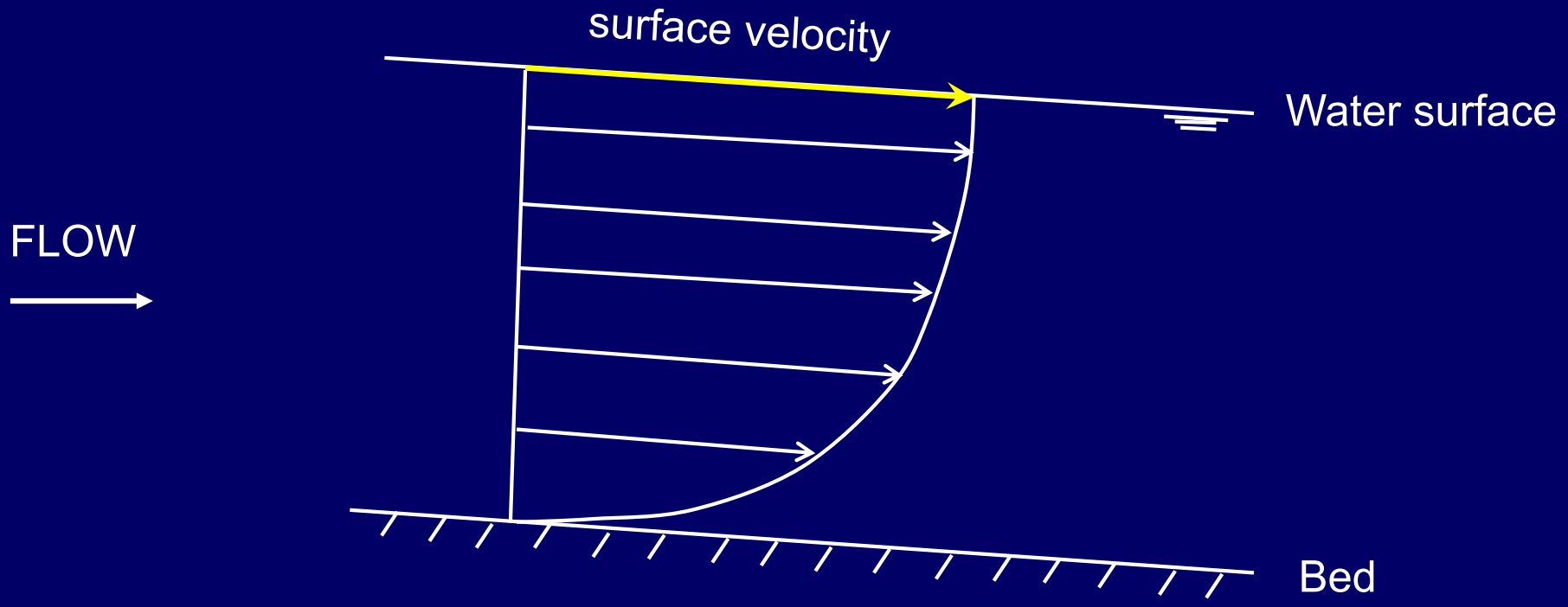
Velocity



Averaging all the local velocities all over the flow cross section
to obtain the cross-section mean flow velocity

Variables of flows: Velocity

Velocity



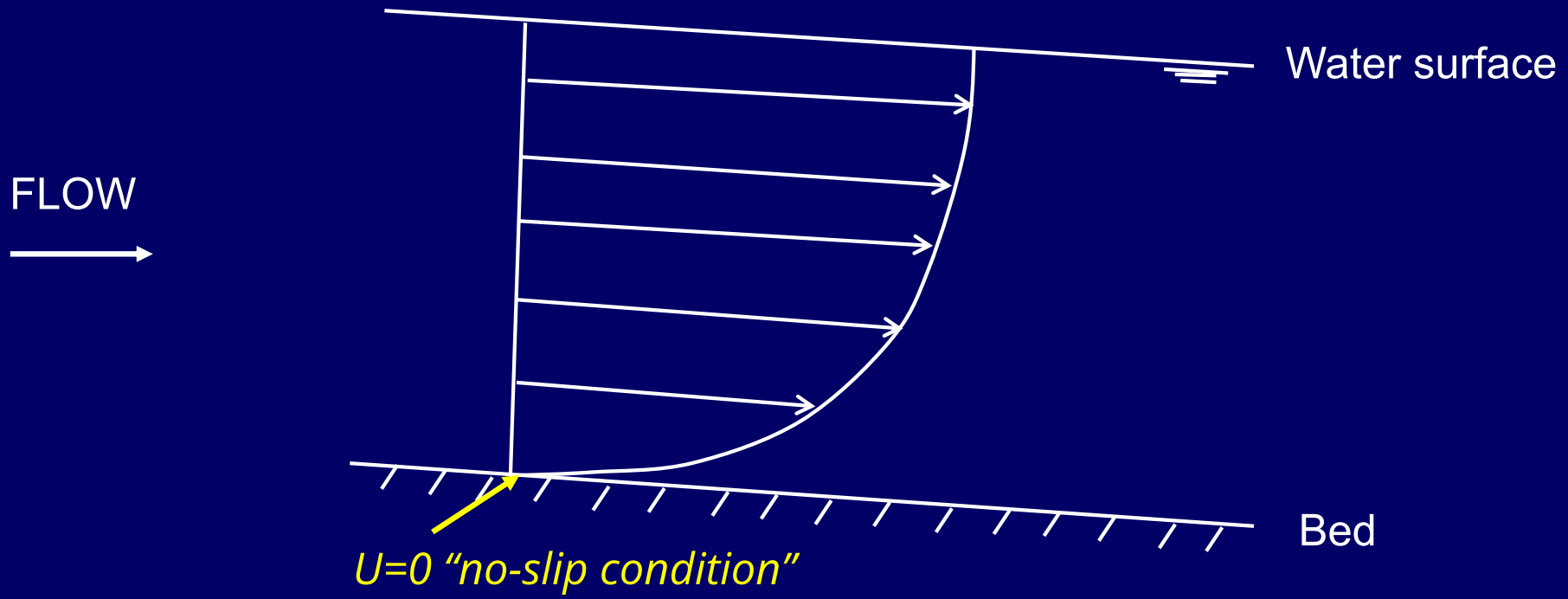
The surface velocity, shown with the velocity distribution in the interior of the flow

Variables of flows: Boundary shear stress

Ways to describe a flow:

- velocity of the flow
- force per unit area the flow exerts on its boundary (boundary shear stress τ_0)

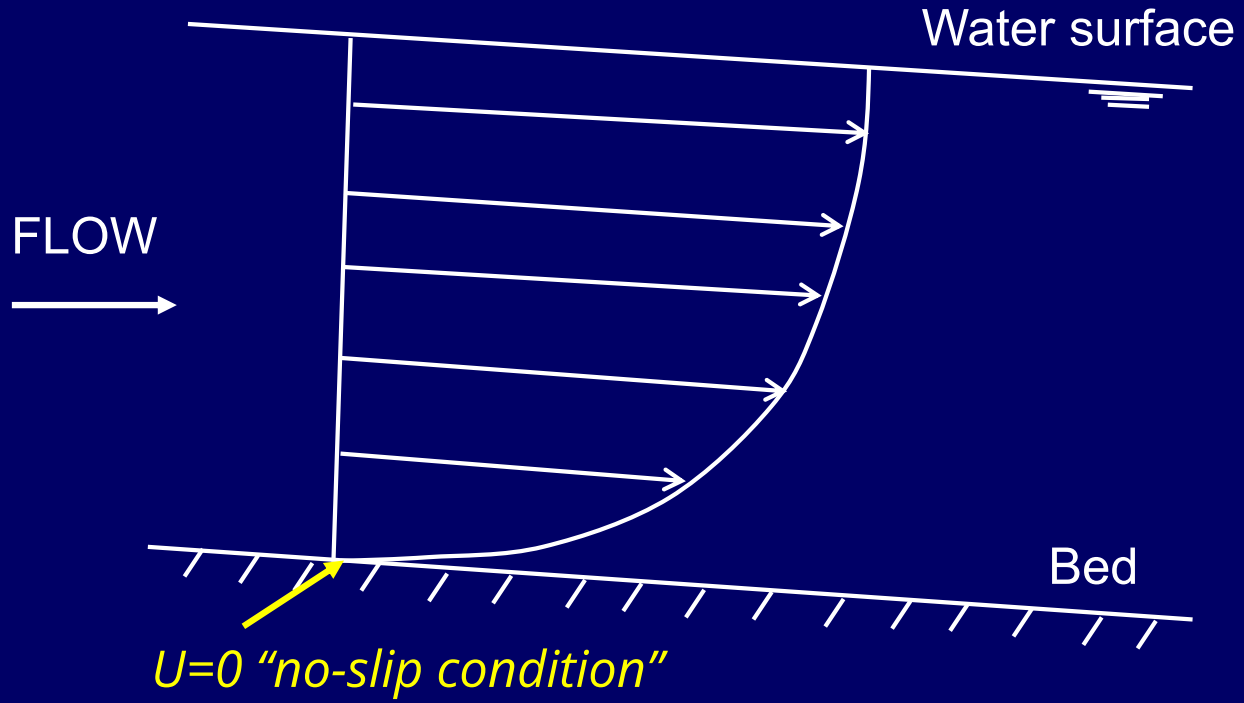
Boundary shear stress (τ_0)



The fluid in direct contact with a solid boundary has exactly the same velocity as that solid boundary

Variables of flows: Boundary shear stress

Boundary shear stress (τ_0)



Shear stress = τ
Proportional to the velocity gradient

$$\tau = \frac{F}{A} = \mu \frac{dU}{dy}$$

Shear stress (resists shearing)

Dynamic viscosity coefficient

Rate of shear deformation (change in velocity with depth)

The fluid in direct contact with a solid boundary has exactly the same velocity as that solid boundary

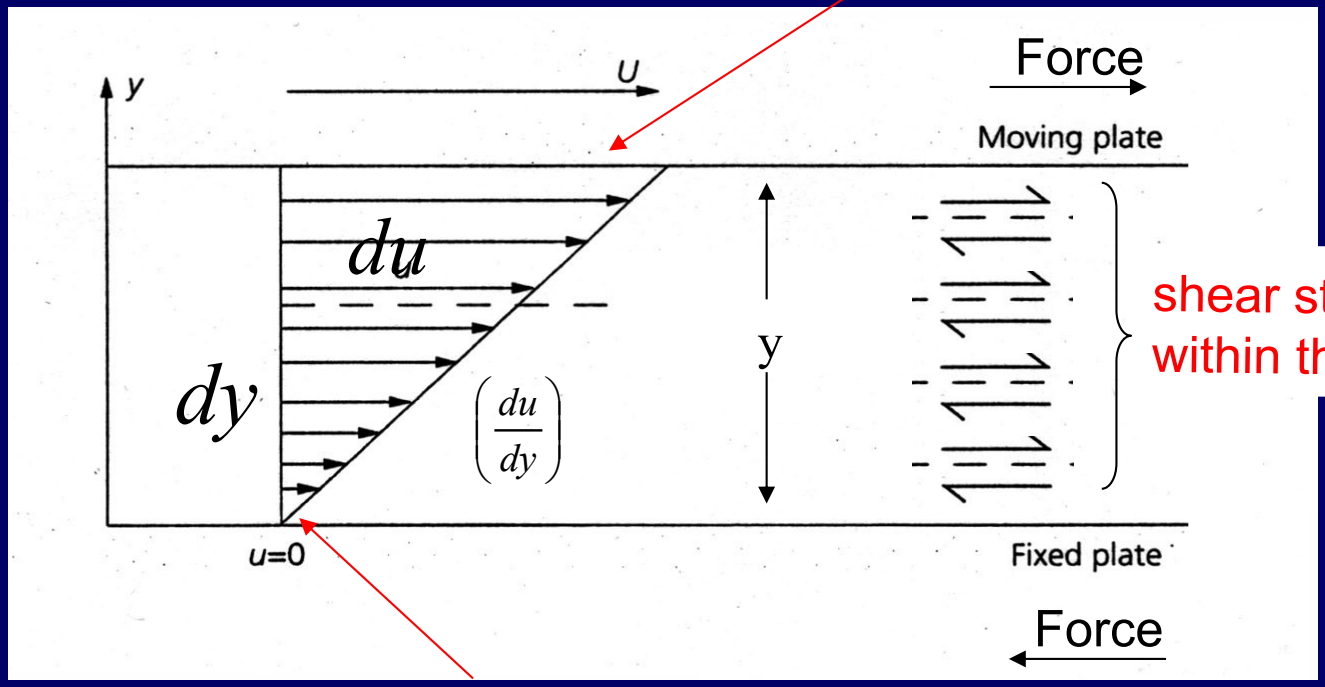
The origin of shear stresses in fluids is the existence of the shearing itself
The stronger the shearing, the stronger the shear stress that's generated
Viscosity is the coefficient that relates the shear stress to the shear in a given fluid

Variables of flows: Visualising dynamic viscosity

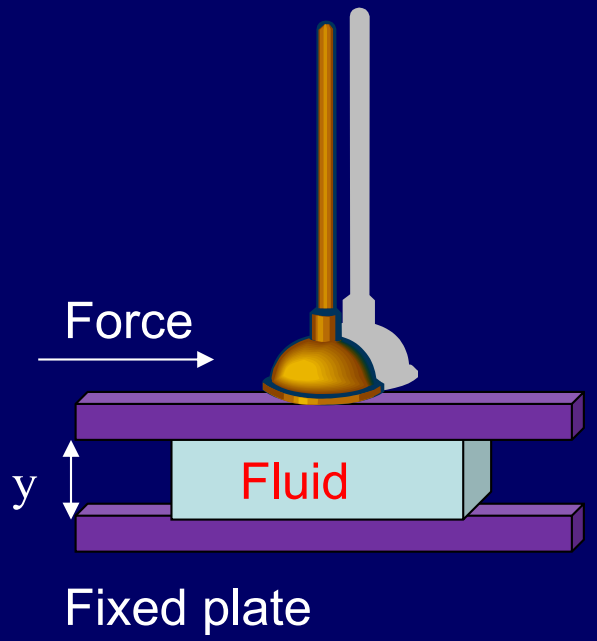
Viscosity is the coefficient that relates the shear stress to the shear in a given fluid

Varies with temperature

$$\tau = \frac{F}{A} = \mu \frac{dU}{dy}$$

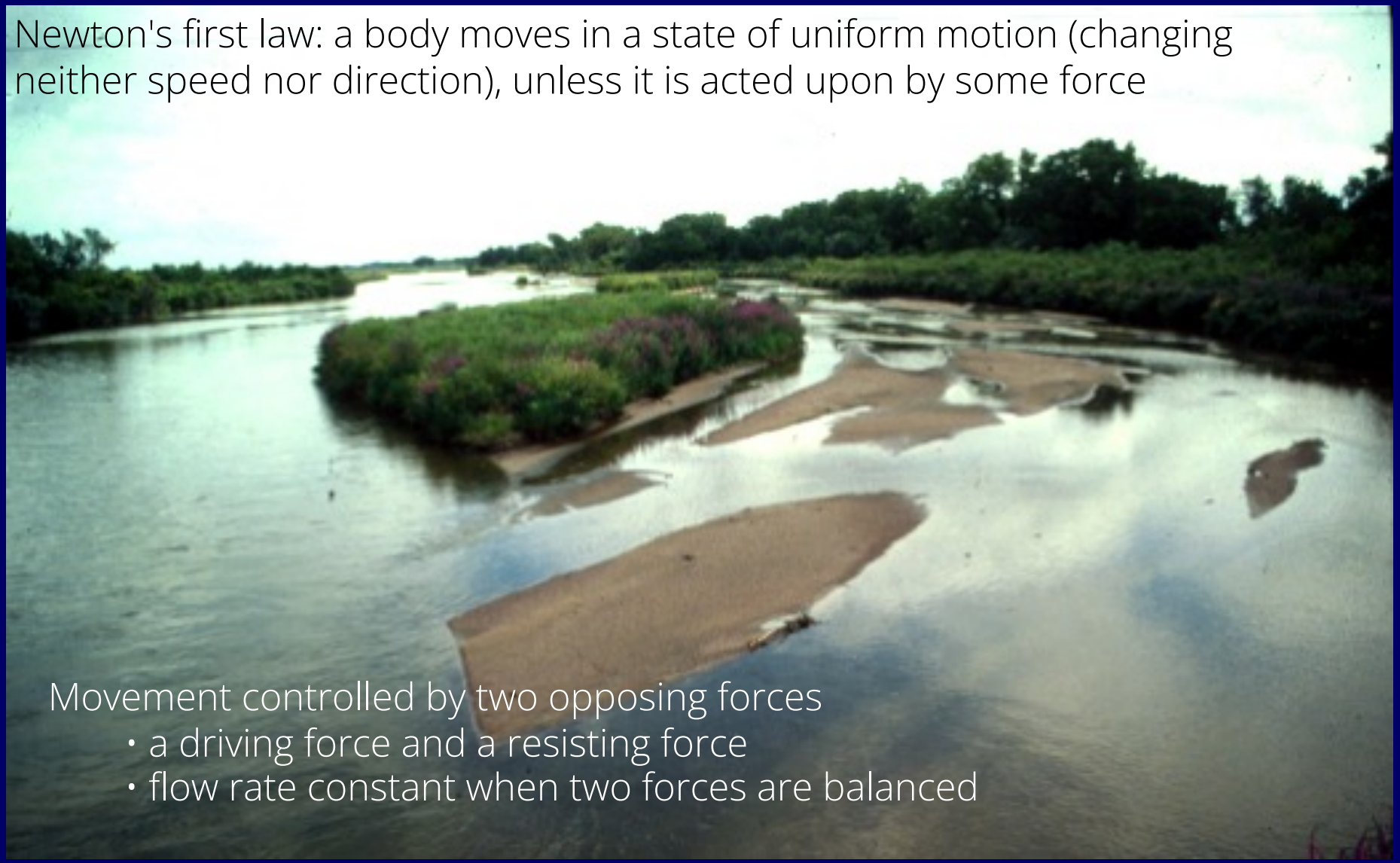


shear stress (τ) within the fluid



Why flows move

Newton's first law: a body moves in a state of uniform motion (changing neither speed nor direction), unless it is acted upon by some force



Movement controlled by two opposing forces

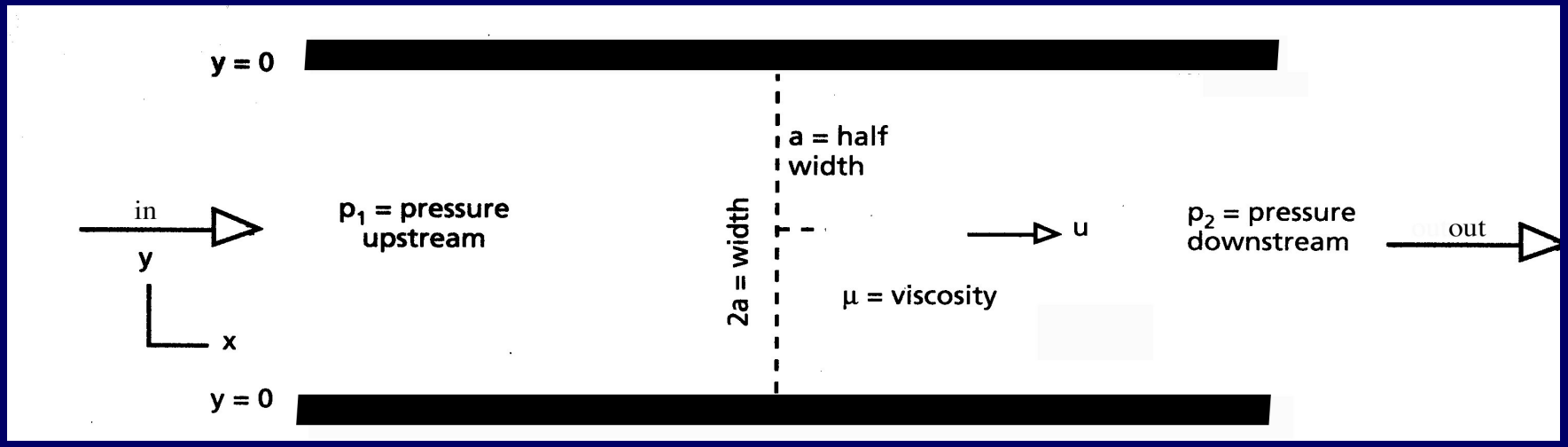
- a driving force and a resisting force
- flow rate constant when two forces are balanced

Platte River, Nebraska

Why flows move: Driving forces I

Driving Forces I: Pressure gradients



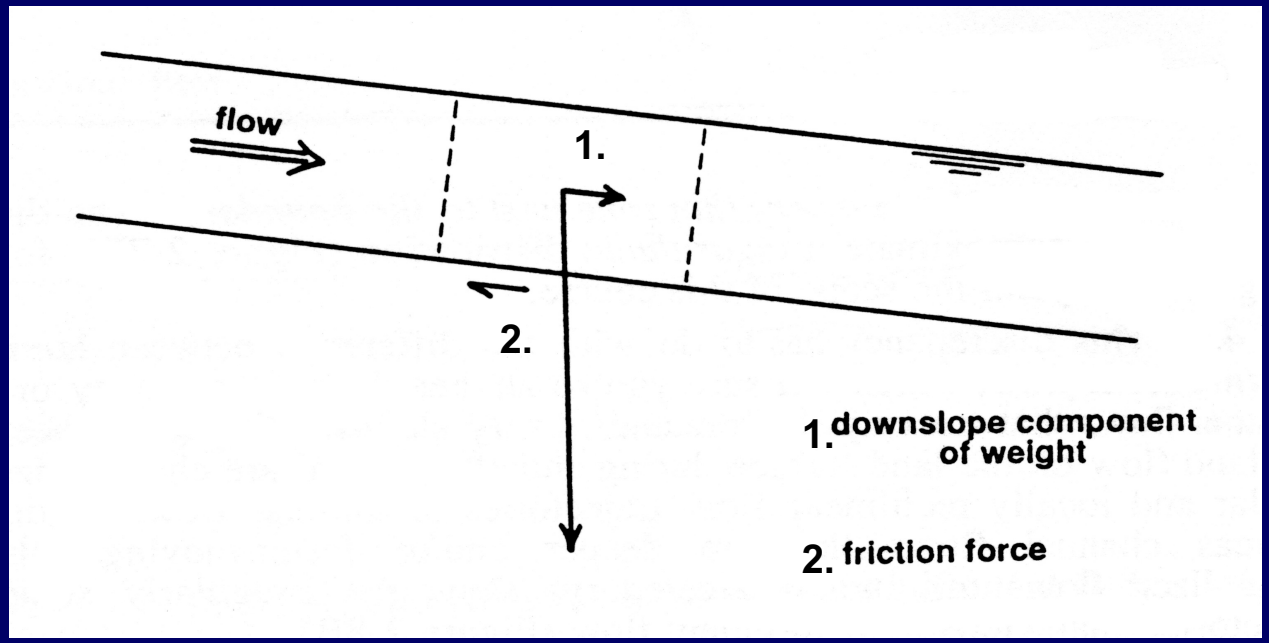


Downstream force counterbalances the frictional resistance exerted on the fluid by the walls of the pipe

Why flows move: Driving forces II

Driving Forces II: The force of gravity





$$\tau = \rho g h S$$

Frictional "resisting" force

Gravitational "driving" force

where:

τ bed shear stress

ρ fluid density

g gravitational acceleration

h depth

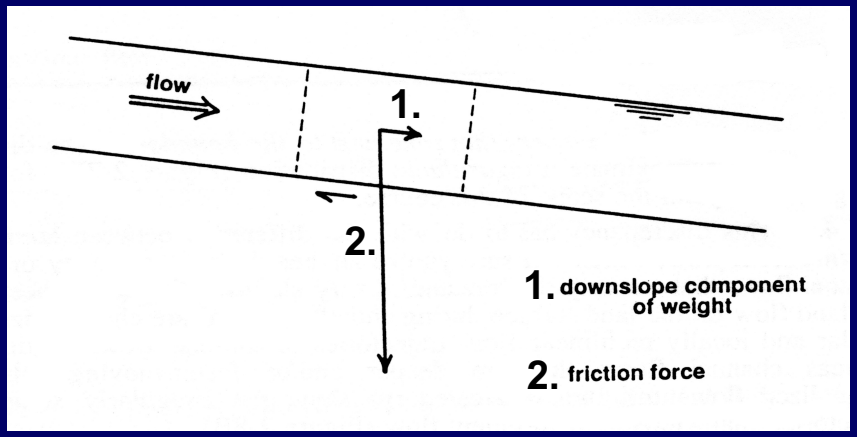
S bed slope

Channel flows



Channel flows





$$\tau = \rho g h S$$

Frictional "resisting" force

Gravitational "driving" force

where:

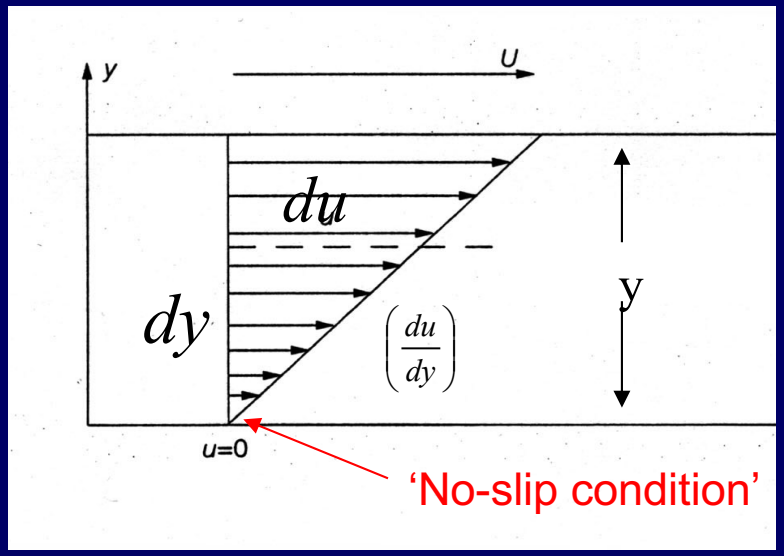
τ bed shear stress

ρ fluid density

g gravitational acceleration

h depth

S bed slope

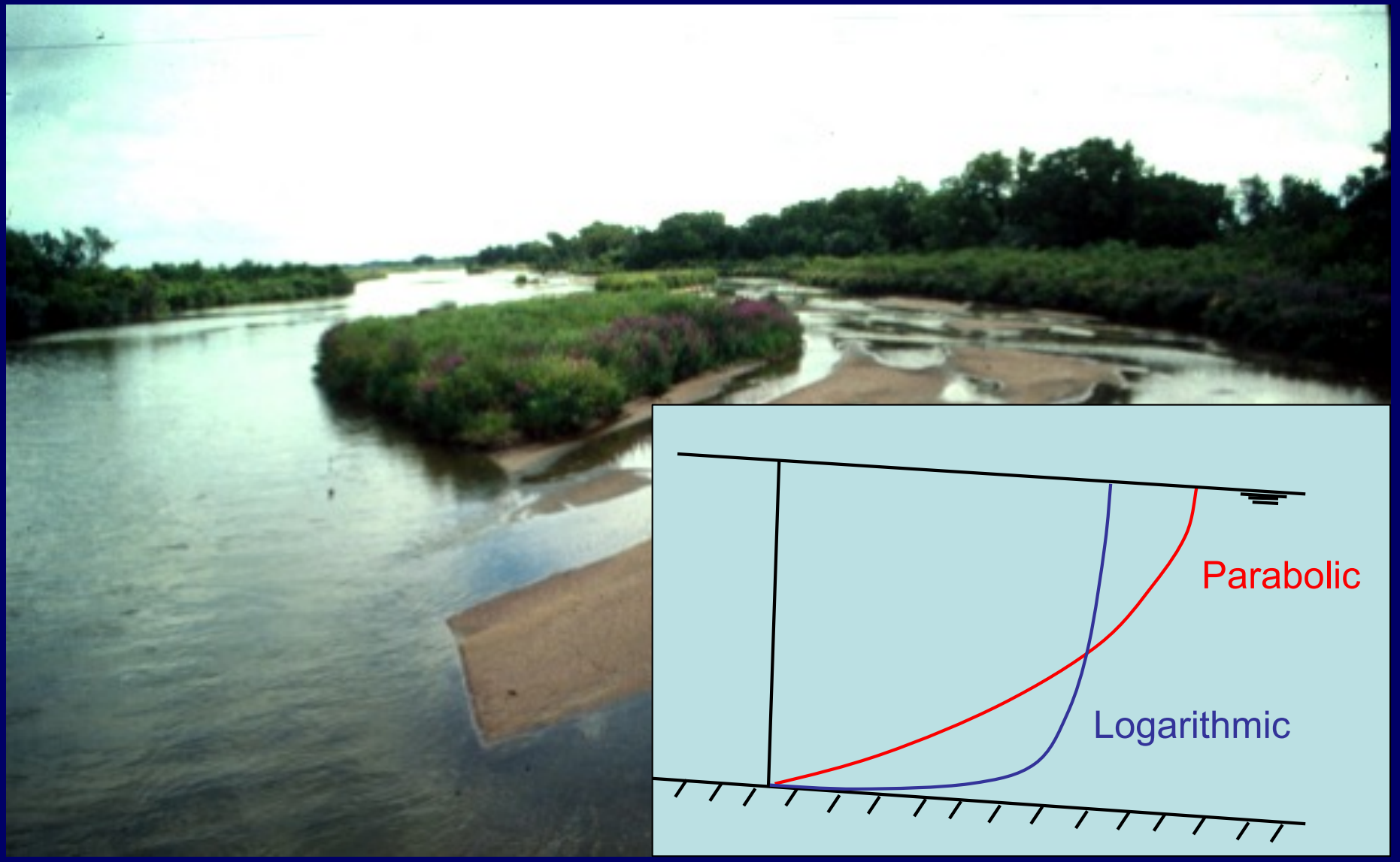


Crude first-order picture of the distribution of velocity through a flow

There is a balance between the down channel pull of gravity on the fluid (1), which acts throughout the flow, and the resistance force the bed and banks exert on the flowing fluid (2)...

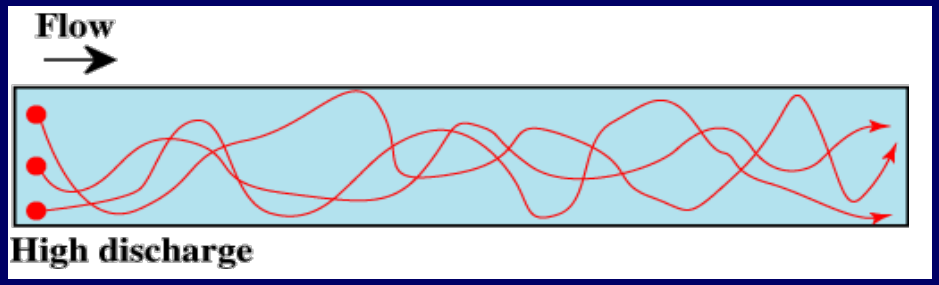
... consequently, the velocity increases upward in the flow, from zero at the boundary, by the no-slip condition, to a maximum at or near the surface

Channel flows

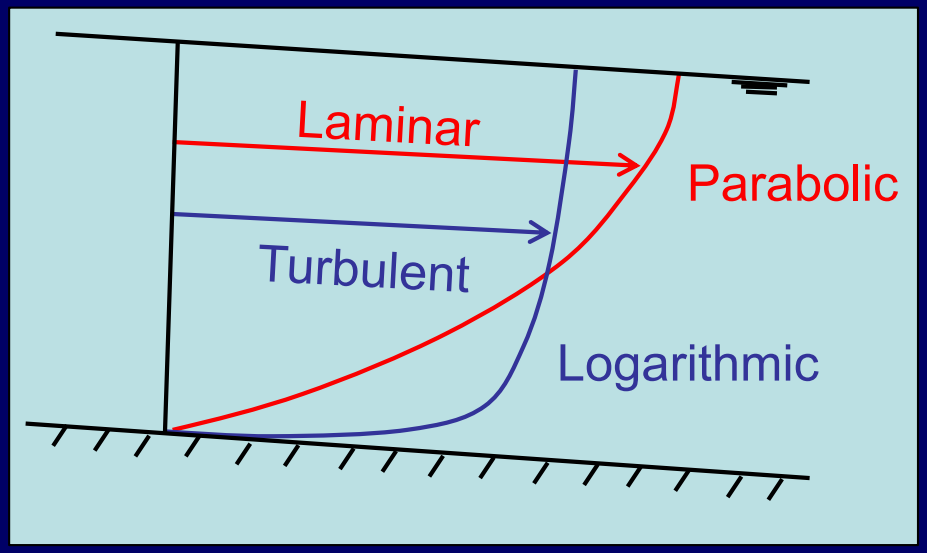


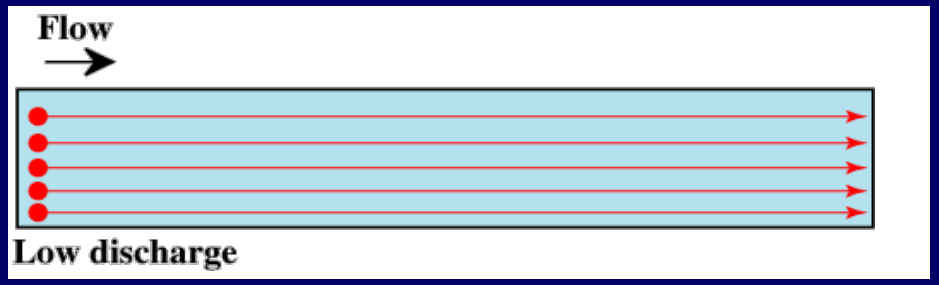
Channel flows





Almost all channelised flows are turbulent





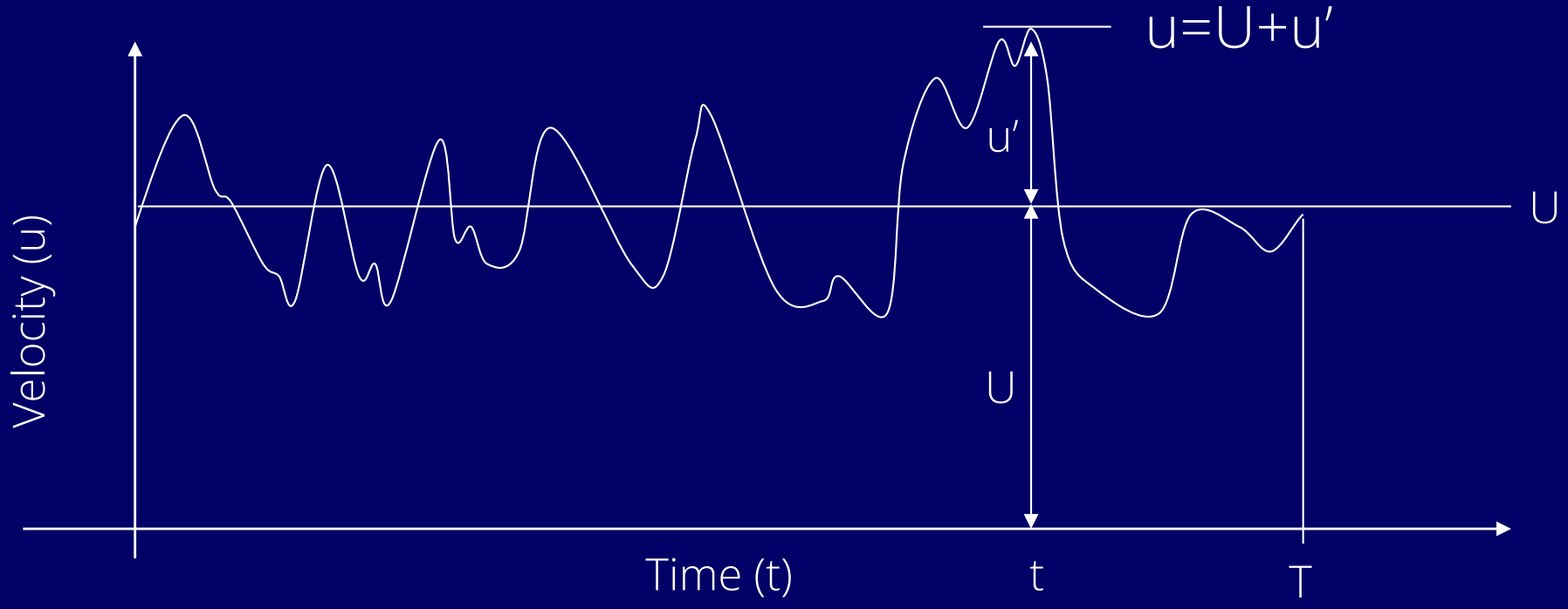
Laminar flow only occurs where flows are either:

- low velocity, or
- very shallow water depths

Visualising turbulence I

Velocity measurements in turbulent flows

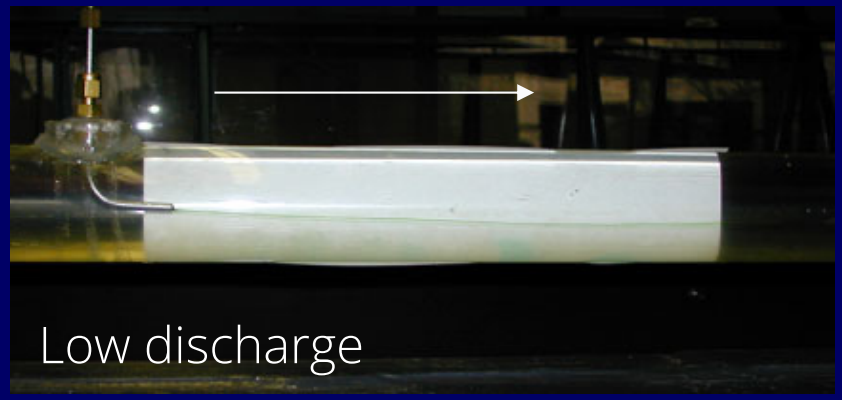
Turbulent flow: a flow in which velocity at-a-point varies in time such that the velocity (u) at any one time (t) is given by the sum of the time averaged mean, U, and the instantaneous deviation from that mean, u' (i.e. u=U+u')



The turbulent fluctuation in velocity, commonly expressed as the standard deviation or root mean square value of the fluctuations, defines the intensity of turbulence.

Visualising turbulence II Dye experiments

Reynolds' experiments involved injecting a dye streak into fluid moving at a constant velocity through a transparent tube







Reynolds' Results

Dye followed a straight path

Laminar flow

Dye followed a wavy path with streak intact

Transitional flow

Dye rapidly mixed through the fluid in the tube

Turbulent flow

Reynolds number (Flow Reynolds number)

Reynolds found that conditions for each flow type depends on:

- velocity of the flow
- diameter of the tube (or flow depth)
- density of the fluid

} Inertial forces (flow inducing/destabilizing forces)

- fluid's dynamic viscosity

} Viscous forces (flow resisting/stabilizing forces- friction)

$$Re = \frac{\rho U h}{\mu}$$

where

ρ	density of fluid
U	velocity of the flow
h	length (tube diameter/flow depth)
μ	fluid's dynamic viscosity

Look at the equation and see that:

- fluids with low viscosity e.g. air, that carry particles in suspension are turbulent
- water flows are only laminar at either low velocities or very shallow water depths
- laminar flow does occur sometimes in debris flows, ice and lava where the viscosities are much higher

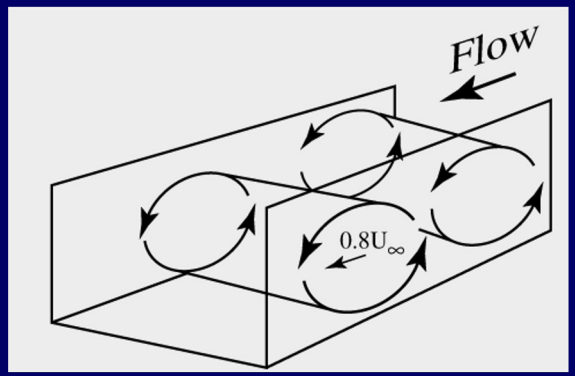
MOST NATRURAL FLOWS OF INTEREST TO SEDIMENTOLOGISTS ARE TURBULENT!!!

Rotational structures within turbulent flows

Eddies



Eddies breaking the water surface and flattening



Components of turbulence that rotate about axes that are perpendicular to the mean flow direction

Vary in diameter from fractions of mm to almost the depth of the flow

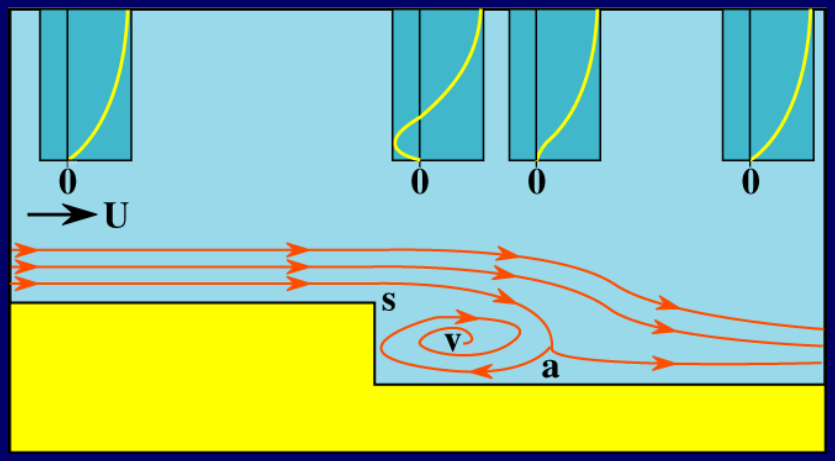
Move downstream with the current at a speed of approximately 80% of the water surface velocity ($U_{¥}$)

Eddies move up and down within the flow as the travel downstream and lead to variation in boundary shear stress over time and along the flow direction

Rotational structures within turbulent flows

Roller eddies

Some eddies are created by the topography of the bed



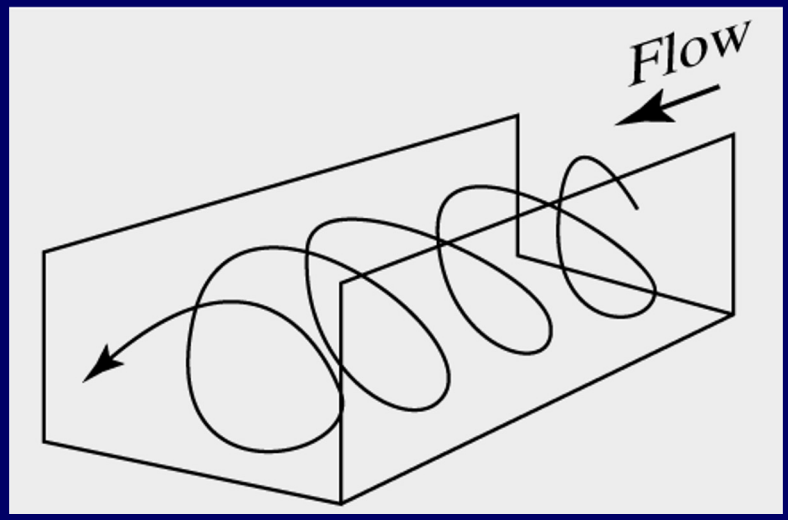
In the lee of a negative step on the bed the flow separates from the boundary ("s" in the figure) and reattaches downstream ("a" in the figure)

A roller eddy develops between the point of separation and the point of attachment

Asymmetric bed forms develop similar eddies

Secondary flows

Large scale rotational structures



Involves a rotating component of the motion of fluid about an axis that is parallel to the mean flow direction

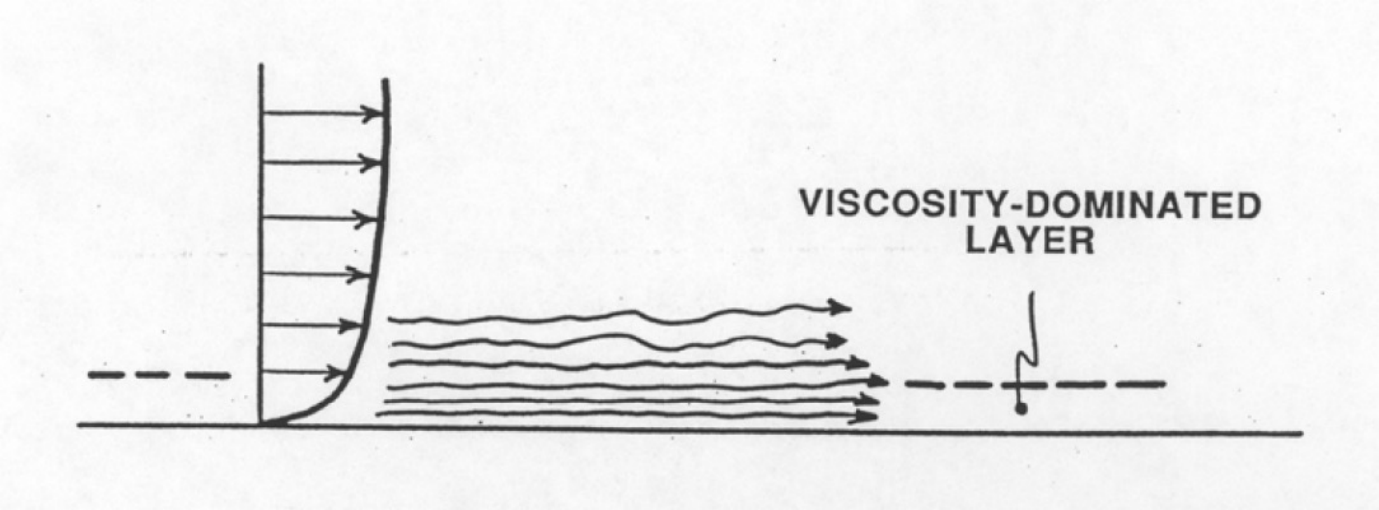
Commonly there are two or more such rotating structures extending parallel to each other

Important in meander bends

Turbulent boundary layers

Key difference between laminar and turbulent flows:

- in turbulent flows the velocity distribution is much more uniform over most of the thickness of the flow, but changes much more sharply very close to the boundary



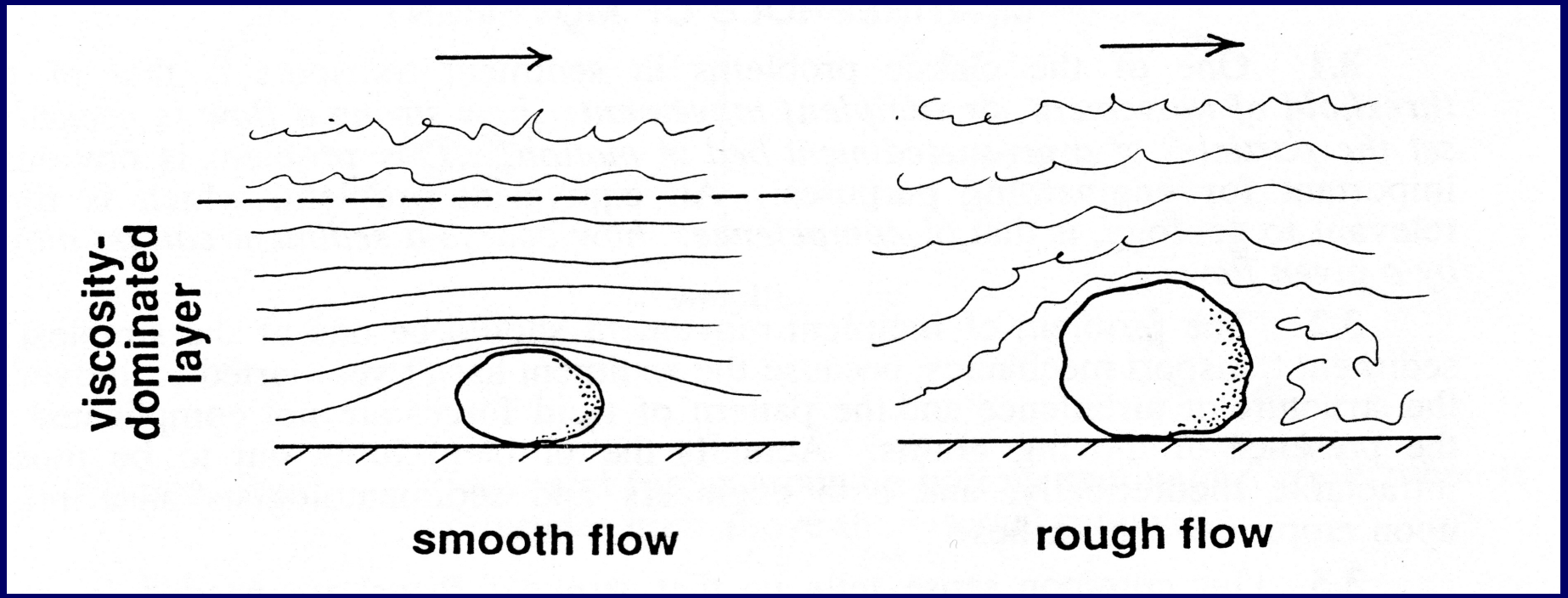
Outer layer: fully turbulent layer

- motion of eddies even out differences in time-average velocity from layer to layer

Viscous sublayer: region near the boundary

- gradient of velocity very steep
  - turbulence can't even out differences in velocity
  - viscous shearing more important than turbulence
- thickness very small % of flow (mms to cms)

Dynamically rough and smooth boundaries

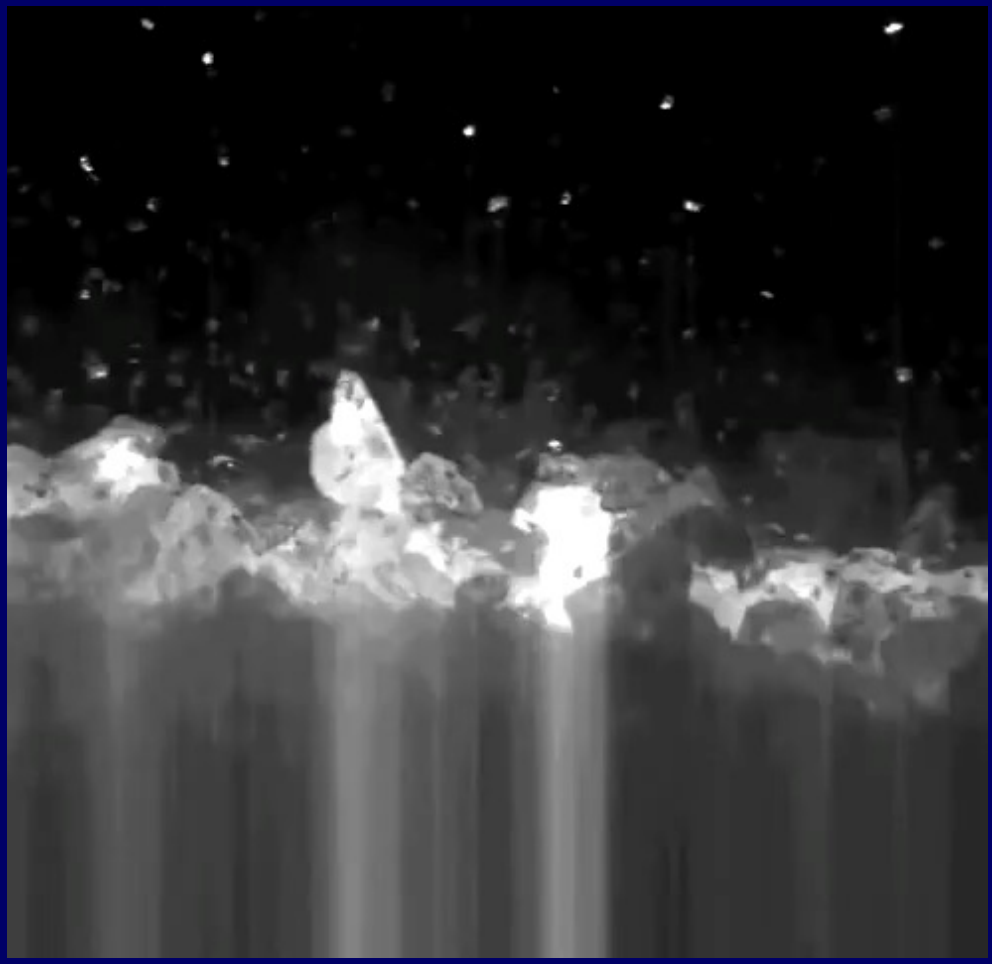


- roughness elements on the bed (sediment particles and/or bed forms) embedded within viscosity-dominated layer

- no overall, continuous viscosity-dominated layer
- viscously-dominated layer only local to the surface of the roughness elements
- roughness elements stick up into turbulent part of flow

Forces on a particle

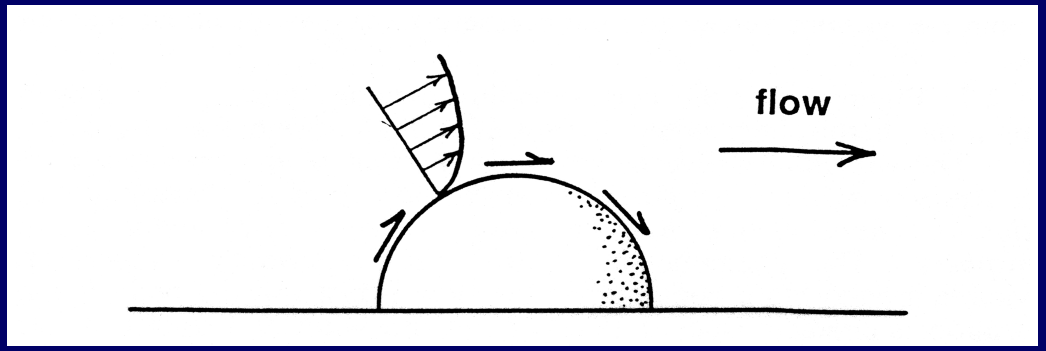
A flow exerts forces on all of the sediment grains exposed at the surface of a loose sediment bed



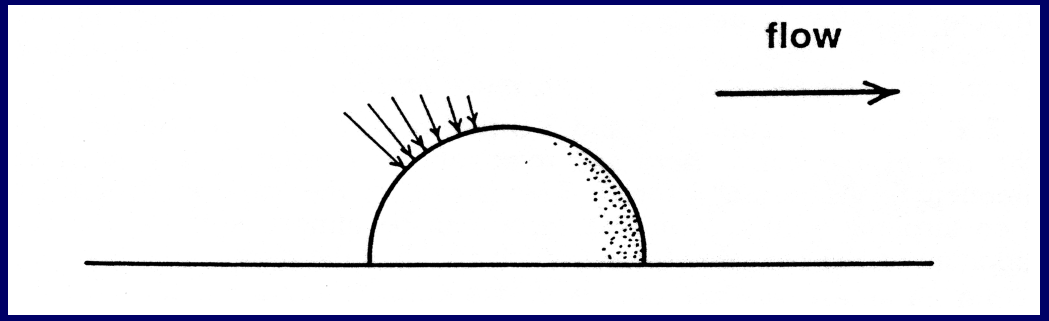
If the force exerted by the flow on a given sediment particle is momentarily great enough, the particle is set into motion

Forces on a particle

Two kinds of forces: viscous forces and pressure forces



Viscous forces
- tangential to the solid surface
- due to shear next to boundary (no slip condition) (surface drag)
- shearing leads to viscous resistance force



Flow around an isolated spherical particle on an otherwise planar boundary

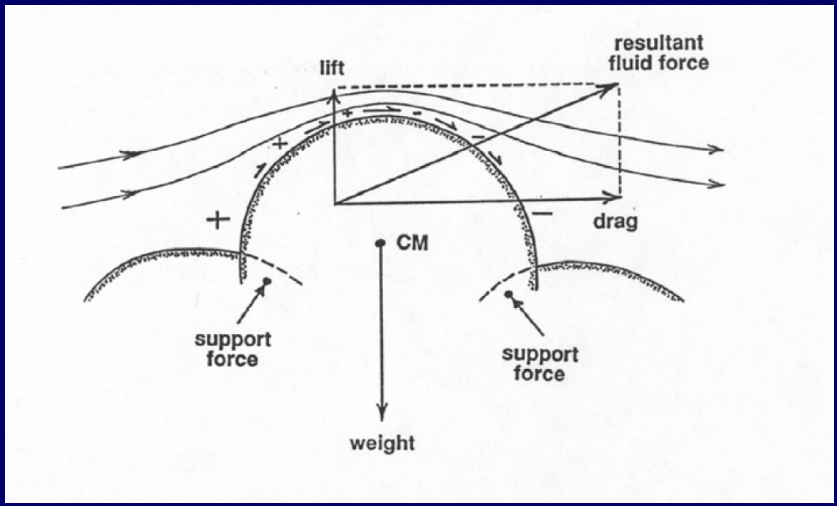
Pressure forces
- perpendicular to boundary
- fluids exert pressure within themselves and on all solid boundaries
  - Hydrostatic (buoyancy)
  - Hydrodynamic (lift and form drag)

Total resultant force on grain surface = viscous forces + pressure forces

Balance of forces on a particle

The nature of the forces on a bed particle depends very strongly on the height of the hemisphere relative to the thickness of the viscosity-dominated layer

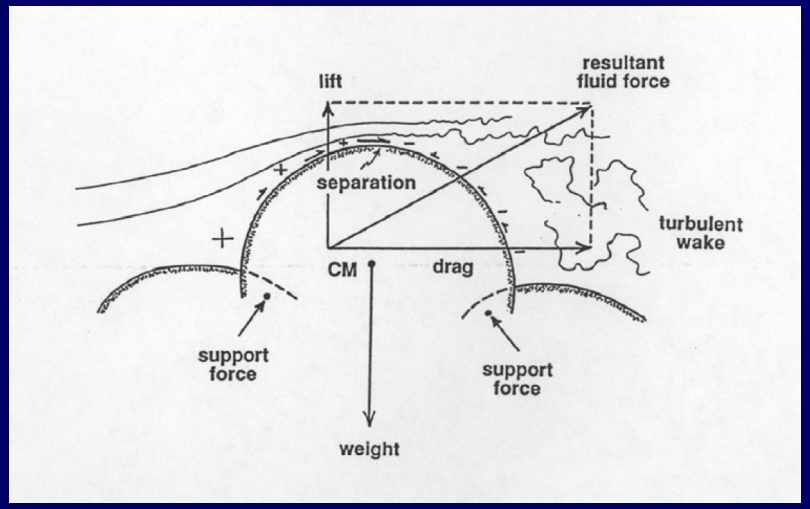
Flow and forces around a bed particle when embedded in viscous sublayer



- streamlines diverge smoothly in front of the hemisphere, converge at the midsection, and diverge again behind the hemisphere (surface drag aka viscous drag aka skin friction drag)
- fluid pressure high upstream and relatively low downstream net downstream pressure force (form drag)

viscous drag + pressure drag

Flow and forces around a bed particle when it protrudes high above the viscous sublayer



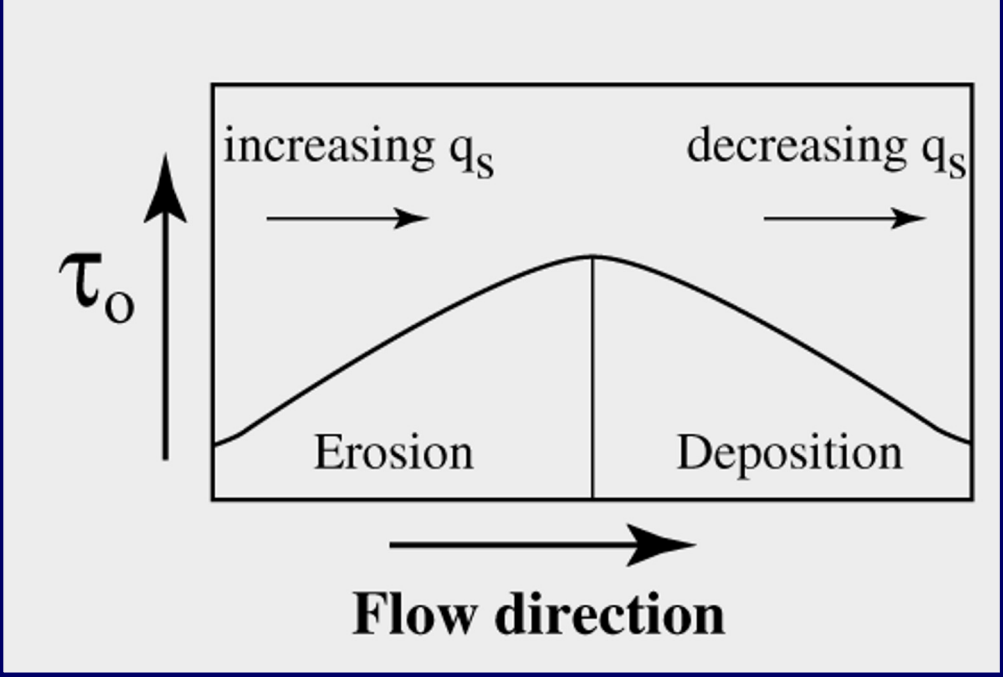
- streamlines diverge smoothly in front then separate from surface in midsection, leaving stagnant region behind the hemisphere (flow separation)
- fluid pressure in stagnant region is low everywhere- net downstream pressure force very large (form drag)

almost all pressure drag

In both cases, the pressure is relatively low at the top of the hemisphere and relatively high around its base, so there's a lift component to the resultant force

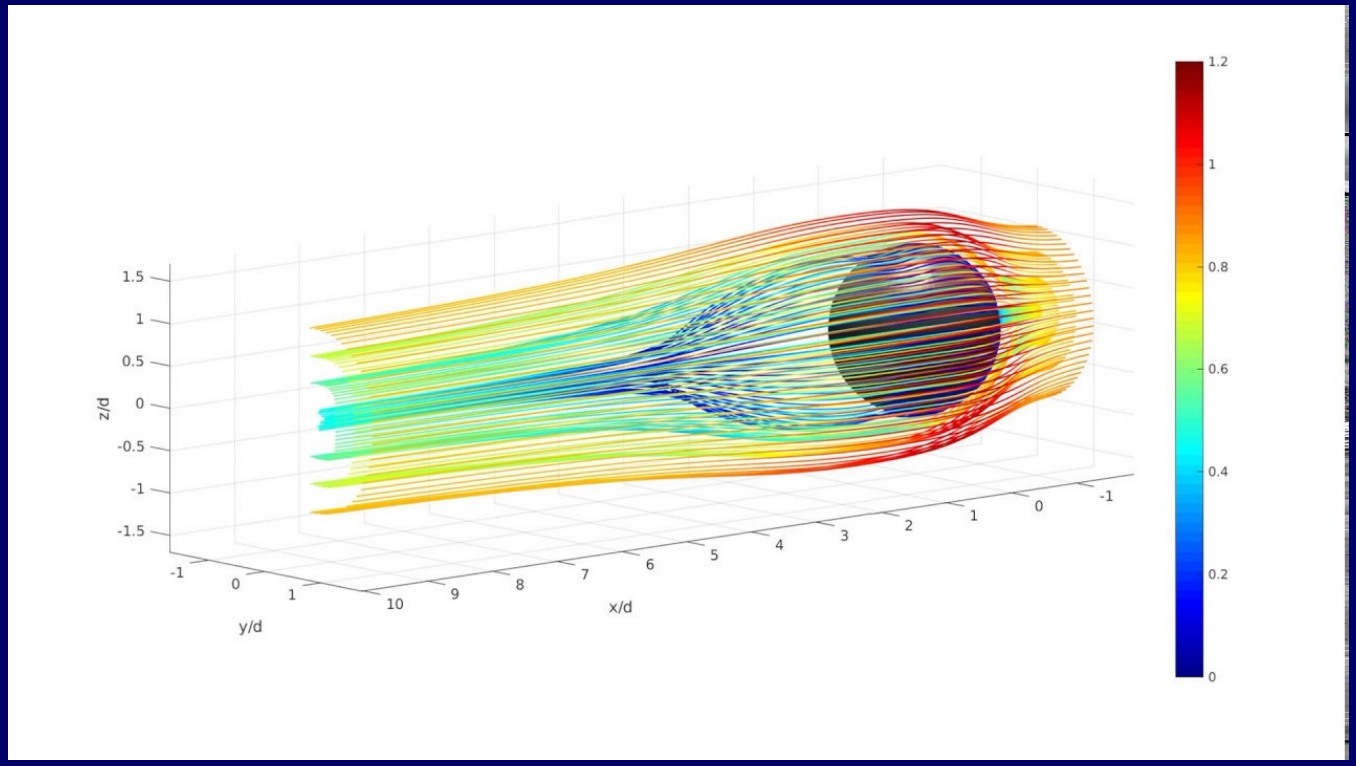
Significance of turbulence for erosion and deposition

- Turbulence results in variation in bed shear stress along and across flows
- Forces can vary by factors of four or even more with time
- Time scales for change in forces on a sediment particle can be very short, fractions of a second
- Bed shear stress governs the power of the current to move sediment; erosion and deposition depend on the change in boundary shear stress in the downstream direction
- In general, sediment transport rate (q_s; the amount of sediment that is moved by a current) increases with increasing bed shear stress



Variation in τ_o along and across the flow due to turbulence leads to a pattern of erosion and deposition on the bed of a mobile sediment

Practical exercise: Particle setting through a fluid



Fluid pressure & Buoyancy



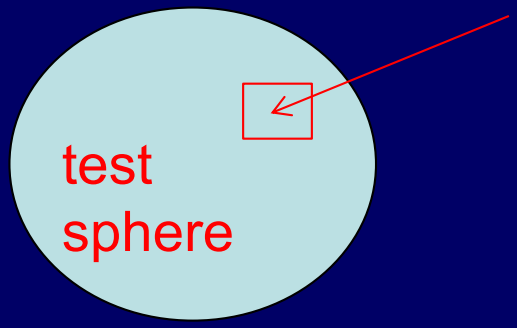
Pressure

the amount of force that is exerted per unit area in a direction perpendicular to the surface

Fluid pressure



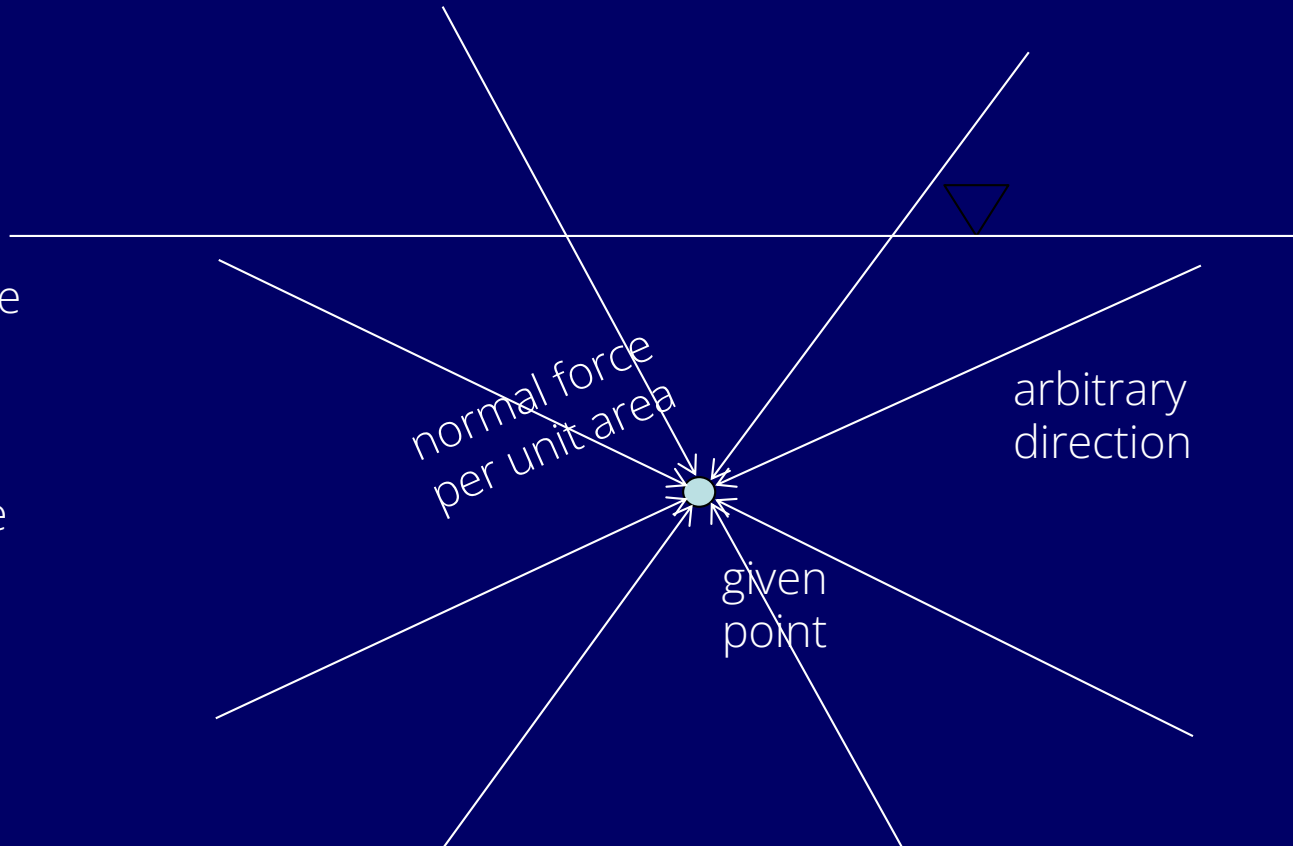
fluid



normal force
per unit area exerted by
fluid at a point on surface
of test sphere immersed
in fluid

= fluid
pressure

Fluid pressure: compressive force per unit area, acts equally in all directions at a point in the fluid (regardless if there is a solid surface at the point on which the force acts)



How big is fluid pressure?

Variation in static pressure with weight of overlying fluid

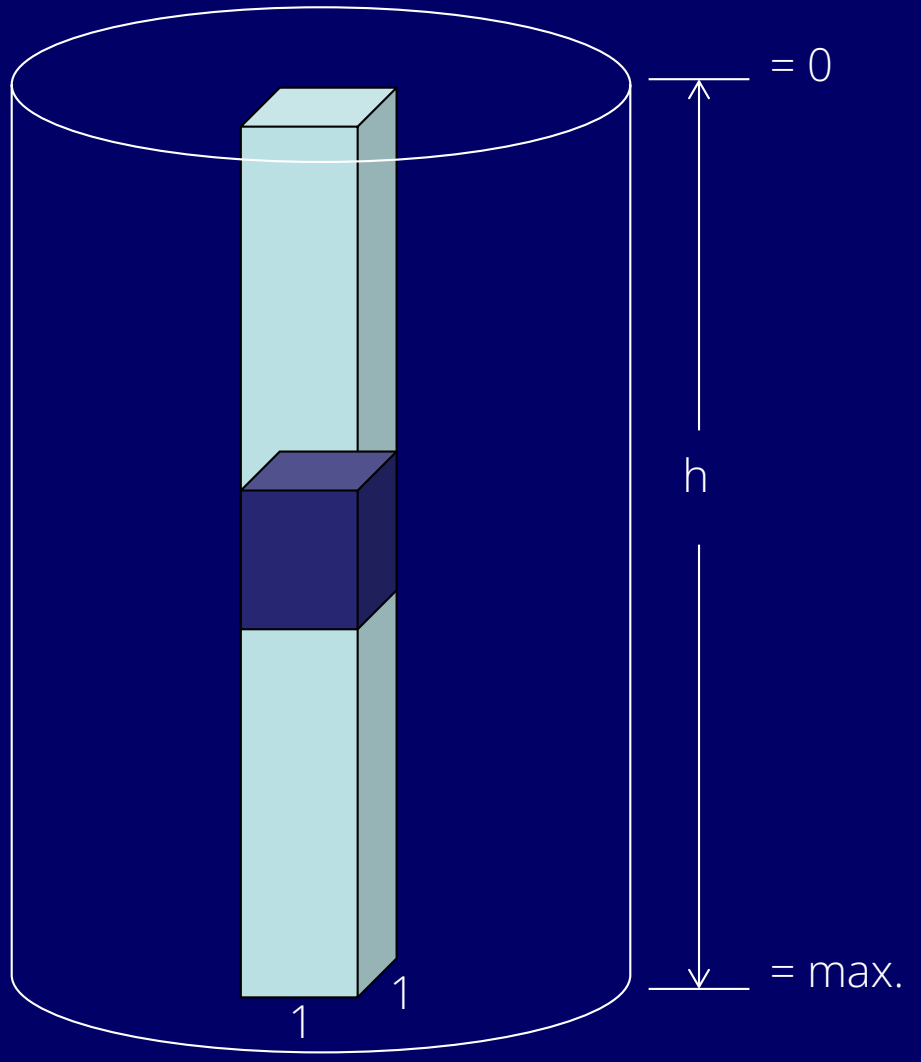
$$p = \rho g h$$

where

ρ	is the density of the fluid
h	height of the liquid column above the bottom
g	is acceleration due to gravity

Hydrostatic pressure: increases linearly with depth

Hydrostatic balance:
upward-directed pressure force is exactly balanced by the weight of the liquid in the body

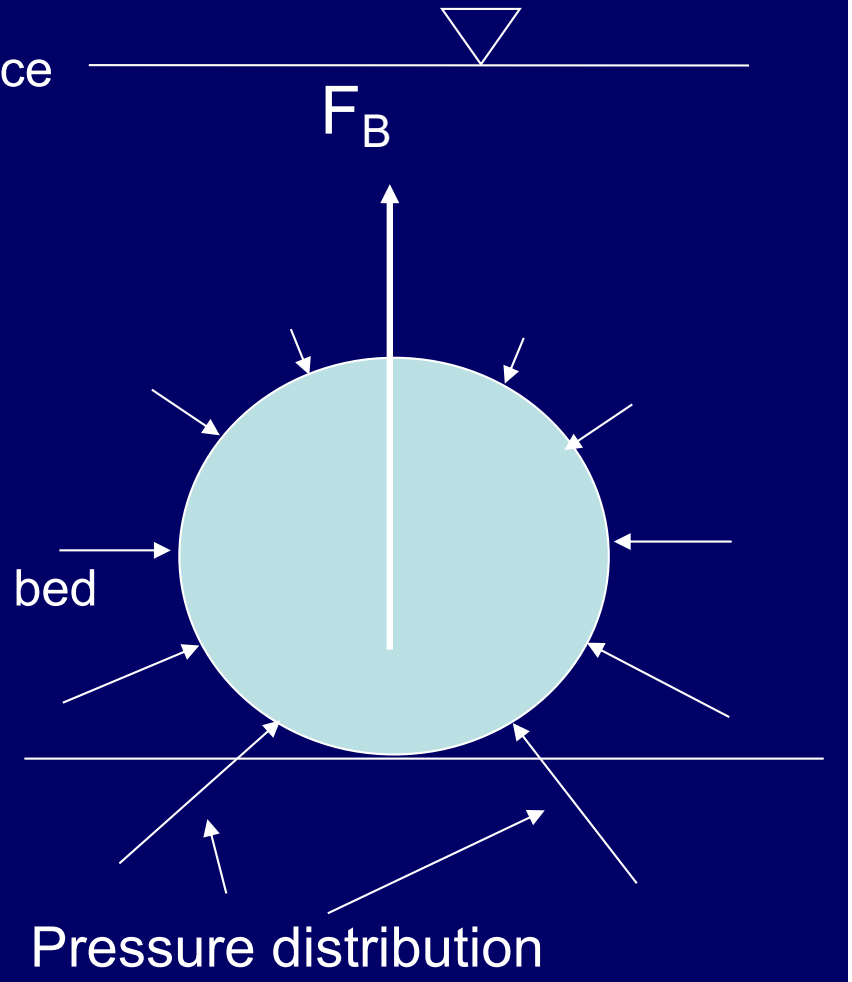


Buoyancy force

$$F_B = -\rho V g$$

where

- ρ is the density of the fluid
- V is the volume of the object submerged
- g is acceleration due to gravity



The submerged specific weight of an object that is entirely immersed in a fluid is less than its actual weight by the weight of fluid it displaces

i.e. specific weight = real weight – weight of fluid displaced